

Nifty100 Financial Intelligence Platform

Analyst Guide

Final operating and interpretation guide

1. Platform Overview

Purpose

The platform combines processed financial statements, analytical KPIs, screening, peer/sector analysis, clustering, reporting and API/dashboard access.

Core layers

ETL and processed data → SQLite → analytics/NLP → FastAPI → Streamlit and PDF/XLSX outputs.

Analyst principle

Use the platform to compare companies consistently while checking data coverage and source quality before interpreting unusual values.

2. Data and Coverage

Universe

The project covers 92 companies in the working Nifty100 dataset.

Datasets

The project integrates multiple financial, market, company, peer and analytical datasets.

Sector coverage

The current processed classification contains 10 broad sectors. This is a documented dataset characteristic; do not fabricate an eleventh sector.

History limitations

ATGL, JIOFIN and SBIN are logged as skipped for the standard three-year tear-sheet policy because sufficient history is unavailable.

3. Company Analysis

Company Profile

Start with the company profile and inspect identity, sector, financial history and available ratios.

Profitability

Review operating margin, return measures and growth together rather than relying on a single KPI.

Balance sheet

Check leverage and capital structure alongside profitability.

Cash flow

Use operating cash flow, free cash flow and capital allocation signals to test earnings quality.

4. Screener Workflow

Step 1

Define the investment question: growth, value, quality, dividend, leverage or turnaround.

Step 2

Use direct KPI thresholds or a predefined preset.

Step 3

Review the returned ranking and inspect the underlying company profile.

Step 4

Treat screening as a shortlist mechanism, not a final investment decision.

Important

Composite scores are most useful when interpreted across the full universe and with awareness of missing-data effects.

5. Peer and Sector Analysis

Peers

Peer groups provide like-for-like comparison where available.

Sectors

Sector views help distinguish company-specific performance from industry structure.

Interpretation

Financial companies, utilities, manufacturers and technology businesses can have structurally different margins, leverage and return profiles. Avoid comparing raw KPIs without context.

Current scope

Use the actual sector labels and counts exposed by the processed dataset and API.

6. Cash Flow Intelligence

Free Cash Flow

FCF is used as a cash-generation signal and should be interpreted alongside operating activity and investment requirements.

Quality

CFO quality and FCF conversion help distinguish accounting earnings from cash realization.

Capital allocation

Review reinvestment, dividends and debt reduction patterns together.

Distress flag

The project includes a rule-based distress signal based on negative operating activity combined with positive financing activity.

7. Clustering and Statistics

Clustering

KMeans groups companies using selected standardized financial features such as growth, FCF growth, ROE, leverage and operating margin.

Use

Clusters are descriptive archetypes, not investment recommendations.

Statistics

Use mean/median profiles, correlations, portfolio statistics and sector outlier reports to understand cross-sectional structure.

Caution

Extreme source values can influence clustering. Investigate data anomalies before treating a cluster label as economically meaningful.

8. Reports, API and Dashboard

Reports

Company tear sheets are in `reports/tearsheets/`; sector reports are in `reports/sector\_reports/`; portfolio summary is in `reports/portfolio/`.

API

FastAPI is served locally on port 8001 and exposes company, screener, portfolio, sector, peer, market-cap, valuation, document and health workflows.

Dashboard

Streamlit provides interactive access and uses the configured API base URL.

Documentation

OpenAPI and Postman artifacts should be kept synchronized with the live API.

9. Data Quality and Final Checklist

Quality checks

Investigate duplicate records, missing history and extreme ratios before interpretation.

Known limitations

The processed sector classification contains 10 broad sectors; three companies are skipped by the standard tear-sheet history policy; some source financial statements contain anomalies requiring review.

Final validation

Run the automated tests, verify key API endpoints, inspect representative PDFs/XLSX outputs, review git status, and commit only intentional changes.

Sign-off principle

Document limitations explicitly instead of hiding them or inventing data.